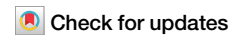


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Heavy metal toxicity and its role as a major driver of past biodiversity crises

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Whether today's heavy metal pollution constitutes an unprecedented threat to biodiversity remains unresolved. Although evidence of metal enrichment exists for several deep-time biotic crises, direct causal links to extinction are lacking. This review synthesizes the current understanding of the relationship between elevated environmental heavy metal concentrations and biological responses, including malformations and extinctions, across marine and terrestrial ecosystems. Drawing on analogies from modern environments, we assess the role of heavy metal toxicity in past mass extinction events. While some studies implicate heavy metals as primary extinction drivers, shown for instance by sporomorphs malformations, others interpret these signals as post-extinction artifacts linked to organic matter decomposition. Our analysis highlights the influence of normalization methods, particularly sulfur-based approaches, and emphasizes the role of metal bioavailability as a key factor influencing biological effects. Further research is needed to clarify the role of heavy metals in Earth's biotic history and ongoing environmental risks.

Human activity has been loading the environment with heavy metals since Roman times¹, and the Industrial Revolution marked a critical turning point in the history of heavy metal pollution, shifting the primary sources of emissions from natural to human-driven activities². Since the 1970s, heavy metals such as mercury (Hg), lead (Pb), and cadmium (Cd) have been acknowledged as a threat to environmental integrity due to their natural abundance, persistence in the environment, and detrimental effects on ecosystems. These inorganic pollutants, characterized by relatively high atomic densities (>4.5–5.0 g/cm³), pose severe risks due to their ability to bioaccumulate in organisms and biomagnify through food chains, even at low concentrations (e.g., European Commission³). As a result, their toxic effects lead to severe damage to body tissues and organs while also causing long-term ecological disruptions, particularly in apex predators and throughout entire food webs^{4,5}. Throughout Earth's history, heavy metal loading was primarily associated with natural processes, including volcanic eruptions and the erosion of metal-bearing rocks. Nevertheless, recent research suggests that heavy metal toxicity has played an important role in past mass extinctions and biodiversity crises, due to the release of vast quantities of heavy metals into the atmosphere associated with Large Igneous Provinces (LIPs)⁶. Some LIPs, such as the Siberian Traps and North Atlantic LIP, have also been linked to contact metamorphism of coal and other carbonaceous sediments that, in addition to carbon dioxide, produce airborne compounds (e.g., coal ash⁷), which contain large quantities of heavy metals that settle or wash out of the atmosphere into oceans, streams, and land^{8–10}. Evidence of heavy metals in the terrestrial and marine

realms has been confirmed by studies in changes in Mercury (Hg) concentrations, normalized Hg concentrations, and isotopes ($\delta^{202}\text{Hg}$, $\Delta^{199}\text{Hg}$, $\Delta^{201}\text{Hg}$), e.g.,^{11–13}. These geochemical proxies serve as a stratigraphical marker and to ground truth the relationship between volcanism and mass extinction events, e.g.,^{12,14,15}. Moreover, heavy metal toxicity has also been invoked as being linked to bolide impacts, such as the Chicxulub impact^{16,17}, which is widely regarded as the cause of the end-Cretaceous mass extinction, e.g.,¹⁸. Given that LIPs and bolide impacts are the most commonly cited triggers of past mass extinction events and biodiversity crises, it is unsurprising that heavy metal toxicity is often hypothesized as a major causal mechanism for a large number of different events from the Early Palaeozoic to the present-day^{19–22}.

The current prevailing model for heavy metal toxicity during mass extinctions suggests that toxic metal emissions during these biotic crises reached poisonous levels, directly contributing to widespread extinctions^{19–21}. Newer models are starting to include the complex interactions of climate change and the methylation of mercury, e.g., through the development of anoxia²². However, these models oversimplify the complexity of extinction dynamics, neglecting: (1) the bioavailability of these heavy metals, which determines their potential toxicity to the organisms; (2) pre-existing environmental stressors, such as increased weathering, collapsed productivity or euxinia, which may have already destabilized/weakened ecosystems resilience prior to metal loading; and (3) the knock-on effects of sulfur emissions and climate change, which interact to introduce additional source of heavy metals in the environment (see Fig. 1). This latter

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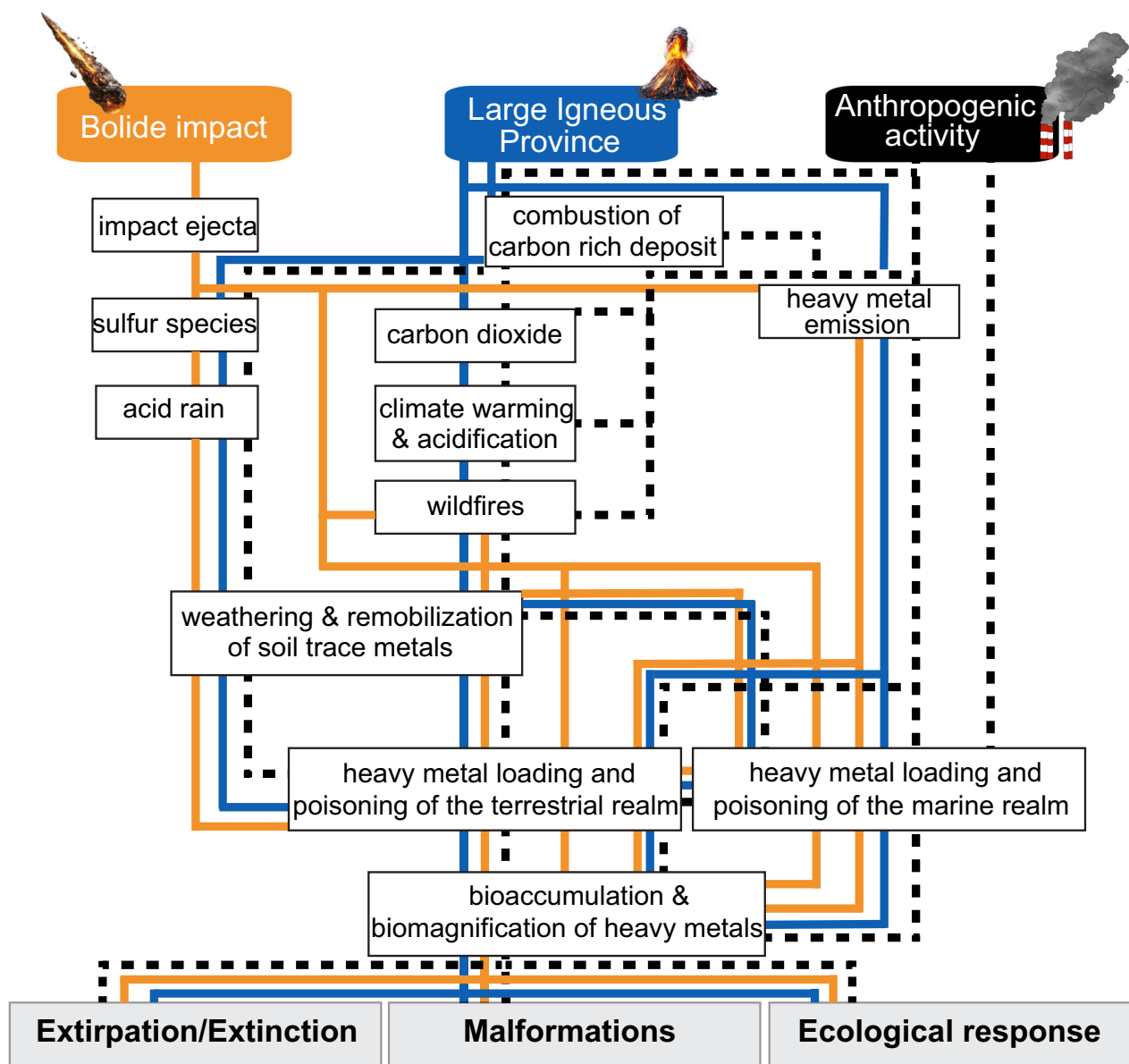


Fig. 1 | Proposed cause-and-effect relationships of bolide impacts, LIPs, and anthropogenic activity with heavy metal toxicity. Despite the different sources of heavy metals, the effects are expected to be similar. Note: Current models do not demonstrate an understanding of this complexity and directly link heavy metal

emissions to toxic metal poisoning. For bioessential metals, thresholds between essential and toxic levels vary across taxa; both depletion and overaccumulation may act as stressors, adding uncertainty to these relationships.

point is supported by preliminary studies from the Toarcian extinction event (~183 Ma) and the Permian-Triassic mass extinction (~252 Ma) suggesting that there is evidence of terrestrial Hg being transported to the ocean^{11,23,24}, further reinforcing the complex link between LIPs, bolide impacts, climate change and heavy metal mobilization within the biosphere.

Here, we review the current state of the literature to test the hypothesis that heavy metal loading of the environment during past mass extinctions and biodiversity crises reached toxic thresholds, potentially contributing to species loss. By doing so, we uncover several key future directions and key outstanding questions for understanding the role of heavy metal toxicity during past biodiversity crises.

Bioaccumulation, biomagnification, and biological impact

Heavy metals pose a dual threat to ecosystems due to their ability to bioaccumulate (higher concentrations in organisms than the surrounding environment) and biomagnify (accumulate to higher concentrations with

increasing trophic levels). For example, mussels can accumulate heavy metals at concentrations 50–100 times greater than ambient seawater, and both consumers and predators higher in the food chain ingest even greater amounts through the consumption of contaminated smaller organisms²⁵. The biological effects of heavy metal exposure can vary: some metals affect biological functions and growth, while others accumulate in one or different organs, causing many serious diseases²⁶. Understanding the role of enriched heavy metals as being toxic or not is also extremely difficult to unravel. This is because certain heavy metals are bioessential elements (aka, physiological metals), meaning they are necessary for biological processes up to a critical threshold that varies between organisms and the heavy metals involved. Beyond this range, overaccumulation can lead to toxicity, while deficiency (often due to malabsorption or environmental limitation) can impair critical cellular functions and increase susceptibility to toxic metal uptake²⁶. Whereas other heavy metals that are not bioessential (aka xenobiotic), such as Hg and Cd, have unknown biological functions and are toxic even at very small concentrations²⁷ (Fig. 2).

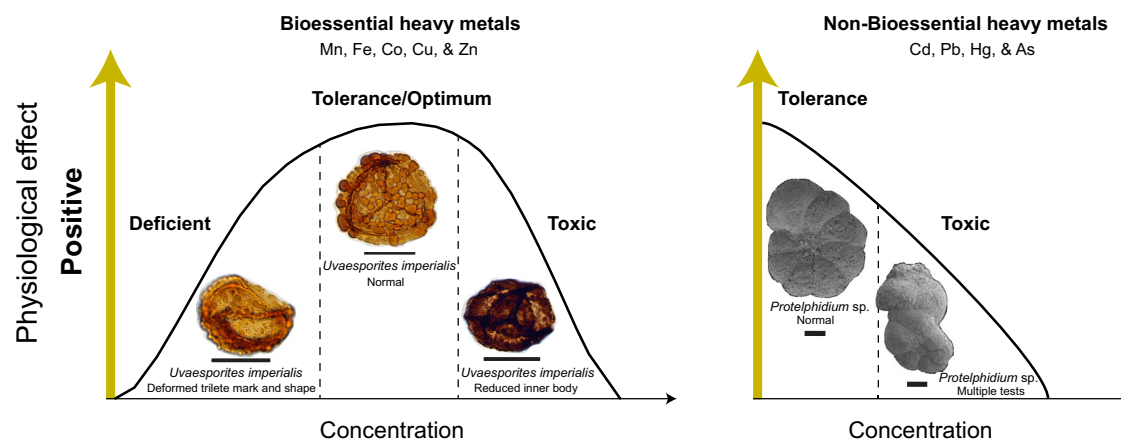


Fig. 2 | Differences in the physiological effects between bioessential and non-bioessential heavy metals in terrestrial sporomorphs and benthic foraminifera. Specimens from Hochuli et al.¹¹¹ (data repository, scale bar 20 μ m) and Ballant and

Carignano⁶² (scale bar 0.050 mm) to show potential morphological responses to elevated heavy metal concentrations.

Heavy metal fractionation and bioavailability

Heavy metal emissions interact with the surrounding environment, which then transform into different metal species and these different species can be divided into different fractions, with differences in their bioavailability. This involves distinguishing 4 phases: (i) Acid-soluble phase—exchangeable metals bound to carbonates that are able to pass easily into the water column, particularly under acidic conditions. It is the fraction with the most labile bond to the sediment and the most dangerous for the environment; (ii) Reducible phase—metals bound to iron and manganese oxides, that can be released if the sediment changes from oxic to anoxic state, potentially triggered, for example, by organisms in the sediments (bioturbation); (iii) Oxidizable phase—associated with organic matter and sulfides, which can become bioavailable under oxidizing conditions. Such conditions can occur, for example, if the sediment is resuspended (e.g., by currents and tides) and the sediment particles come into contact with oxygen-rich water; (iv) Residual phase—lithogenic and inert metals that are tightly bound in mineral structures and are virtually non-bioavailable²⁸. For example, with Hg, identifiable fractions include (i) Exchangeable (Hg_{ADS1}), (ii) Bound to organic matter (Hg_{ABS}) and Hg sulfide (HgS), (iii) Mercury sulfate and oxide (Hg_{ADS2}), and (iv) Residual (Hg_{RES}), where the labile fractions (Hg_{ADS1} , Hg_{ABS} , and Hg_{ADS2}) can be easily released from sediment and readily transformed in the environment, whereas HgS and Hg_{RES} are insoluble and incorporated into mineral lattices and not biologically available²⁹. Previous studies have also shown that marine organisms can intercept and absorb the labile (easily mobilized) fractions, making them bioavailable and potentially hazardous to food webs²⁹. In deep-time studies of heavy metal abundances, the concentration of Hg is often normalized against TOC (Hg/TOC) or sulfur (Hg/S) to account for the variation in organic-matter (OM) and sulfides drawdown, e.g.,^{30–32}, while other heavy metals are often normalized against Al. However, these normalizations do not account for changes in metal bioavailability or fractionation, which are crucial for assessing environmental toxicity. Recent research demonstrates that heavy metal fractionation provides deeper insights into past environmental toxicity. Mercury fractionation, in particular, has been successfully applied to deep-time sedimentary records³³, showing promise for reconstructing the bioavailability and ecological impacts of heavy metals during past hyperthermal events.

Heavy metal toxicity and its role in environmental crises

While mercury has been extensively studied in the context of past environmental crises, largely due to its application as a volcanism proxy, the roles of other detrimental heavy metals like copper (Cu), chromium (Cr), cadmium (Cd), lead (Pb), arsenic (As), cobalt (Co), and nickel (Ni) have received comparatively less attention^{34–36}. These heavy metals are typically

removed from the water column by binding to organic matter or forming insoluble compounds with sulfides and subsequently incorporated into sediments. As a result, their elevated concentrations in the sedimentary record can, in part, be attributed to fluctuation in the amount of total organic carbon (TOC) or development of euxinic (anoxic and sulfide-rich) conditions¹⁴. However, such interpretation has led to controversy about whether the elevated heavy metals were a direct driver of past extinctions or simply a byproduct of broader geochemical change e.g.,³⁷. Even when heavy metals are normalized against TOC, aluminum (Al) or titanium (Ti), uncertainties persist due to the complex effects of taphonomic processes on geochemical signatures. For example, the development of mercury sulfides in the sediment column or the impact of burial-related thermal maturity are poorly understood^{32,38}. In marine environments, mercury can be sequestered either within organic matter or as insoluble mercury sulfides, the latter requiring euxinic conditions to develop³⁹. In such cases, it is unclear whether Hg toxicity added substantially to ecosystem collapse, or whether euxinia was sufficient to drive extinction events on its own. This uncertainty underscores the need to re-evaluate how mercury enrichments are interpreted in the sedimentary record, particularly during intervals of extreme environmental change. Due to Hg's strong affinity for organic matter e.g.,^{40–42}, accurate interpretations of Hg anomalies require evaluation of TOC-normalized concentrations^{43,44} to avoid misinterpretations 'created' through variations in organic matter. A considerable constraint of the Hg/TOC approach lies in its sensitivity to samples containing low TOC concentrations (<0.2 wt%), where the analytical uncertainty can undermine the reliability of enrichment calculations^{12,45}.

Comparisons of Hg/TOC trends across five major hyperthermal events that coincided with biotic crises: the Permian-Triassic transition, Carnian Pluvial Event, Toarcian Event, latest Maastrichtian Warming Event, and the Palaeocene Eocene Thermal Maximum (Fig. 3) show broadly similar patterns of elevated Hg/TOC, suggesting a potential link to enhanced volcanic activity and/or widespread ecosystem disruption. In most studies investigating Hg enrichment trends, Hg concentrations are normalized to TOC (see Fig. 3). While some studies have taken the threshold of low TOC (<0.2 wt%) into account^{13,46,47}, others have placed less emphasis on it^{23,34}. For consistency across datasets, and because raw TOC values were not reported in some cases²³, this cutoff was not applied in our comparative analysis, but samples with TOC below the threshold were highlighted in red (Fig. 3). The latter suggests that the earliest Hg enrichments identified for the Permian-Triassic transition³⁴, might not be a reliable signal, but instead from a methodological artifact associated with a low TOC content. Similarly, for the latest Maastrichtian Warming Event, most of the data falls below 0.2 wt % TOC⁴⁷, indicating that the use of Hg/TOC to determine Hg enrichment is likely unsuitable for this interval. A different normalization method would

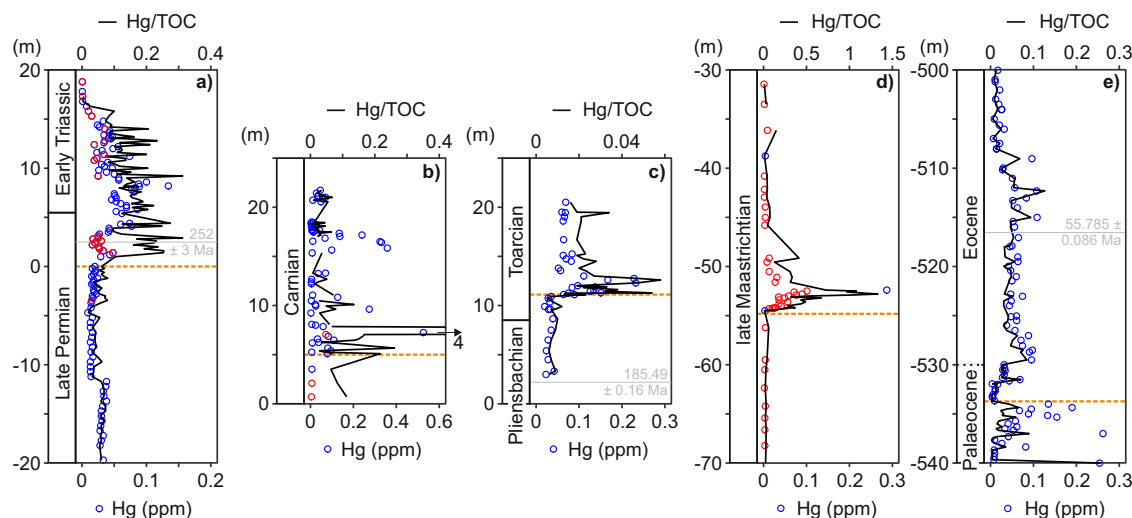


Fig. 3 | Mercury concentration (ppm) and Hg/TOC (ppm/wt%) data across selected extinction and hyperthermal events. a Permian-Triassic transition, Festningen, Svalbard³, **(b)** Carnian Pluvial episode, Steinbach and Polzenberg sections, Austria⁴⁶, **(c)** Toarcian event, Bighorn Creek, Alberta, Canada²³, **(d)** latest

Maastrichtian Warming event, Jiaolai Basin, China⁴⁷, and **(e)** Paleocene-Eocene Thermal Maximum, Central Basin, Svalbard¹³. The orange dotted line marks the onset of the carbon isotope excursion in each interval and red circles were used for samples, which had a TOC concentration under the suggested threshold of 0.2 wt%.

be more appropriate, in particular as Hg accumulation in sediments is also influenced by additional environmental and depositional factors beyond organic carbon content, including variations in lithology, detrital input, and redox conditions, particularly the presence of euxinia. To disentangle these effects, additional normalization of Hg concentrations to detrital tracers such as aluminum (Al), thorium (Th), or titanium (Ti), and indicators for euxinia, such as sulfur (S), is required. Despite the importance of multi-proxy normalization, only⁴⁷ reported and applied both Al- and S-normalized Hg data, allowing for a more robust assessment of whether observed Hg enrichments reflect increased Hg flux or are driven by other depositional and/or geochemical conditions. Grasby et al.³⁴ provided Al and S data, but did not apply these for normalization of Hg, while the studies by Jones et al.¹³ Them et al.²³ and Mazaheri-Johari et al.⁴⁶ lacked sufficient data to assess detrital or redox influences.

While mercury leads the discussion of metal toxicity in past biotic crises, this review advances the field by examining the roles of other biologically toxic heavy metals such as As, Pb, and Cd. Through two case studies from different depositional environments during the Permian-Triassic Mass Extinction (PTME), we provide a broader perspective on multiple heavy metals (Fig. 4a, b).

Case Study 1: Heavy metal enrichment in a siliciclastic section during the Permian Triassic mass extinction

At the Festningen section (Svalbard), application of sulfur-normalized Hg data considerably alters the interpretation of previously identified Hg enrichments based solely on Hg/TOC³⁴. Throughout the Early Triassic, Hg concentrations appear elevated; however, when normalized to S, these enrichments disappear, suggesting Hg accumulation in the sediment corresponds to increased sulfide content (see Fig. 4a). The development of euxinia and associated sulfide formation is further supported by elevated pyrite concentrations reported by Grasby et al.³⁴, suggesting that the apparent Hg enrichment reflects sequestration of Hg as insoluble mercury sulfides under euxinic conditions, rather than from a true increase in Hg flux. As mercury sulfides are not bioavailable, they are unlikely to have exerted a toxic effect on marine organisms. Instead, the anoxic, hydrogen sulfide-rich conditions would have posed a more immediate threat to marine ecosystems. It is worth noting that both the development of euxinic conditions and mercury enrichments in the sediment occur after the extinction horizon (loss of bioturbated sediment and silica production) at Festningen.

A similar pattern is observed for Cd, which shows a flat trend when normalized to S, further indicating that redox conditions, rather than external inputs, have played a major role in its distribution (Fig. 4a). In contrast, As and Pb remain elevated even after normalization to S near the extinction horizon (± 5 m). When further normalized to Al, their enrichments appear independent of a lithological control, implying a potential increase in biologically relevant toxic metal concentrations during the extinction interval. Nevertheless, without direct bioavailability proxies, their ecological significance cannot be determined based on the bulk sediment data alone.

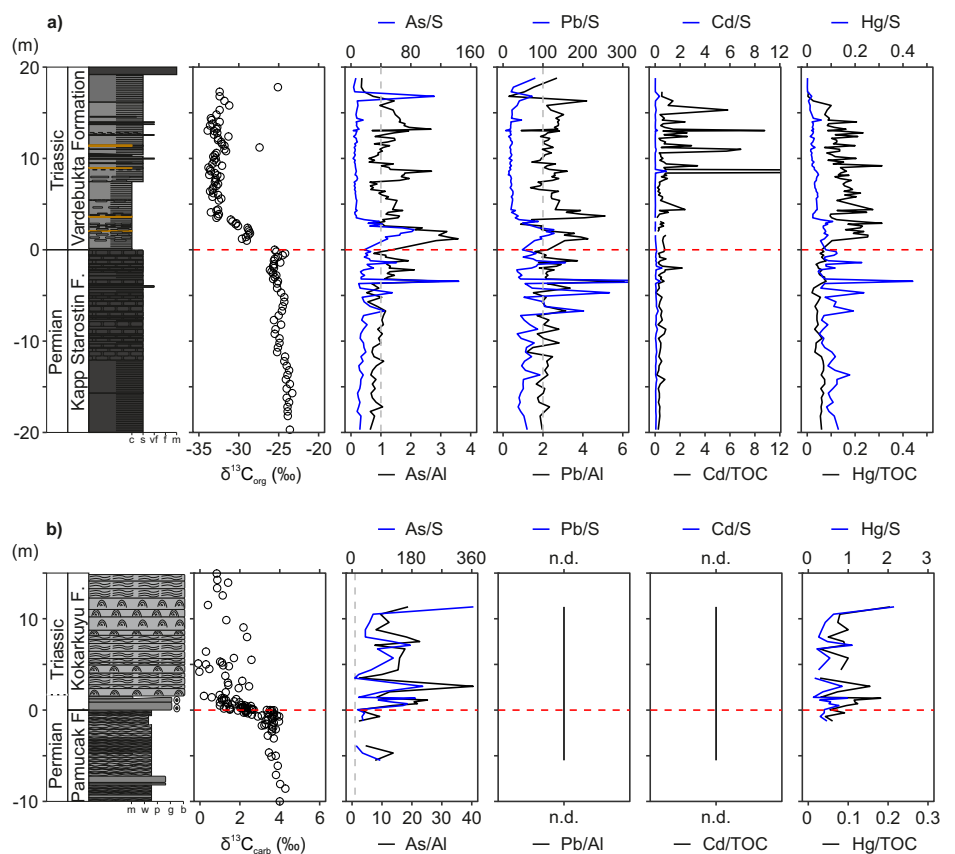
Case Study 2: Heavy metal enrichment in a carbonate succession during the Permian Triassic mass extinction

In contrast, Çürük Dağ (Türkiye), located within a carbonate platform, shows low or undetectable Pb and Cd concentrations, suggesting no substantial increase in sedimentary fluxes for these metals around the extinction horizon (Fig. 4b). However, both As and Hg show variable trends across the section, regardless of the normalization method. When normalized to Al and S, these elements appear considerably enriched compared to Festningen. Yet, this discrepancy is likely a product of differing baseline levels, as the Al and S concentrations at Çürük Dağ are approximately one order of magnitude lower than at Festningen. This highlights a critical methodological challenge as comparisons between sections with differing lithologies, or even within heterogeneous sections, must be approached with caution, since baseline differences can inflate normalized values. One potential solution would be the use of enrichment factors calculated against standard materials or background values. Currently, this approach is readily applied using Al as the baseline, thereby accounting only for lithological variation, but comparable methods using S or TOC are not yet widely adopted. Even Al-based enrichment factors have recently been shown to be biased toward low-Al samples, producing a similar effect to that seen in the Hg/TOC proxy when comparing samples with different lithologies. While an adjusted method for calculating Al-based enrichments has been proposed to address the bias, these findings underscore that methodologies for identifying metal enrichments are still evolving. Publicly available datasets should therefore be critically evaluated using state-of-the-art methods before being applied^{48,49}.

Biological response of marine ecosystems

Despite the well-known detrimental consequences of elevated heavy metal concentrations in modern marine ecosystems, there are no studies (beyond

Fig. 4 | $\delta^{13}\text{C}$ and heavy metal trends across the Permian–Triassic mass extinction event. **a** the siliciclastic Festningen section, Svalbard³⁴ and **(b)** carbonate platform section, Çürük Dağ, Türkiye ($\delta^{13}\text{C}$ from Demirtaş¹¹⁷ and Richoz¹¹⁸; major and trace element data from Frank¹¹⁹). As and Pb concentrations were normalized to S and Al (ppm/wt %), while Cd and Hg concentrations were normalized to S and TOC (ppm/wt%). The Pb and Cd concentrations from Çürük Dağ were consistently below the detection limit of 3 and 0.3 ppm, respectively. The red dashed line represents the extinction horizon at both sections.



marine palynology) that directly investigate the role of heavy metal emissions in driving mass extinctions and biodiversity crises in the marine realm. “Direct studies” are meant here as studies that show a link between heavy metal toxicity and extinction beyond a temporal relationship. An exception is the observed asymmetric geographical response of nannoplankton during the end-Cretaceous mass extinction, interpreted as a consequence of heightened heavy metal toxicity in the Northern Hemisphere¹⁷. Studies on the effects of anthropogenically induced heavy metal pollution have shown that the impacts are recorded in surviving species, in the form of morphological malformations, e.g.,⁵⁰, structural changes in the communities, e.g.,⁵¹, incorporation of heavy metals into skeletal material, e.g.,^{52,53}, and the accumulation of heavy metals into animal tissues, e.g.,⁵⁴.

Malformations

Morphological abnormalities in marine organisms have been intensely studied in modern environments. It is important to highlight that abnormalities can be divided into two types: deformations/discontinuous abnormalities (i.e., developed following mechanical damage to an organism) and malformations/continuous abnormalities (i.e., developed during the growth of an organism). Evidence based on modern studies suggests that heavy metal pollution would produce malformations as it affects the growth of organisms rather than result in physical damage. Examples of heavy metal induced malformations include: bivalves with both posterior and ventral scores, flattening of the posterior shell, changes in height/thickness ratios and asymmetry, with up to 46–87.5% of specimens showing malformations in heavily polluted areas^{50,55,56}; gastropods with shorter spires, globular shape, smaller relative size, globular malformations on the inner surface⁵⁷; and benthic foraminifera, with reduced chamber size, twisted and distorted chamber arrangement, overdeveloped chambers, aberrant chamber shape, changes in coiling direction, and malformations can even be so extreme that species identification is not possible^{58,59}. The intense investigation of benthic foraminiferal malformations in modern ecosystems has also led to the

establishment of the Foraminiferal Abnormality Index (FAI), which has shown up to 23% of communities showing malformations in heavily polluted areas⁶⁰.

Such morphological malformations are also preserved in the fossil record and have been recorded at the end-Cretaceous mass extinction in foraminifera^{61,62}, the end-Triassic mass extinction in ammonites⁶³, end-Permian mass extinction in foraminifera and ammonoids^{64,65}, and the Frasnian-Famennian extinction in conodonts and brachiopods^{66,67}. In addition, certain nannoplankton and planktic foraminifera described from the end-Cretaceous mass extinction and Eocene hyperthermals have also been described as “Excursion taxa”, characterized by malformations that deviate from the typical symmetrical structures, often showing features such as weak or excessive calcification, e.g.,⁶⁸. There are even cases where species are defined by their growth malformations and likely represent an extinction survivor of another species, e.g., *Guembelitra irregularis*, a planktic foraminifera from the end-Cretaceous event⁶¹.

Morphological abnormalities have been heavily investigated in modern, coastal marine ecosystems; however, because multiple environmental factors can cause malformations, previous studies have found it difficult to isolate exactly which environmental changes, or specific heavy metal(s), are the cause of the malformations^{58,69}. One key reason why previous studies have not combined malformation development with independent metal fractionation data is that the use of malformations was developed as a cost-effective and efficient biomonitoring tool, especially for developing nations, e.g.,^{70,71}. A similar problem exists for deep-time occurrences of malformations, which means that even though malformations have been reported from the fossil record, scientists have indicated that malformations can be associated with heavy metal toxicity but have held back in directly suggesting that heavy metal toxicity played a role in the past, e.g.,^{61,62,64}. This means that both in modern and past studies, the utility of malformations, without simultaneous tests for the bioavailability of heavy metals from the same samples, makes malformations as a bioindicator of heavy metal toxicity equivocal.

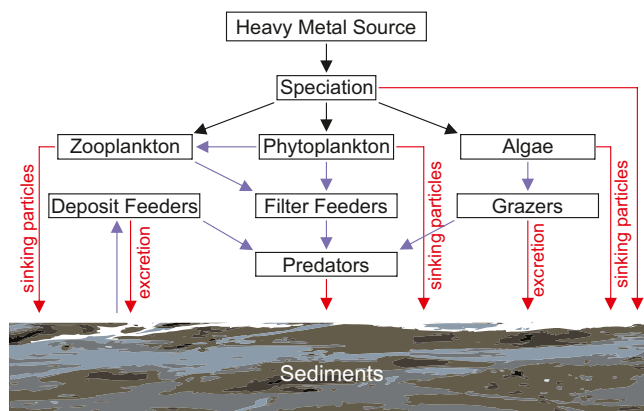


Fig. 5 | Transfer and bioaccumulation (purple arrows) of heavy metals through a marine ecosystem and their relationship to the ecology of different organisms. Note: bioturbation, erosion and water column conditions can remobilize heavy metals from the sediment but are not included in the diagram.

Similar to deep time studies, challenges exist when understanding historical ecological responses to heavy metal pollution, because it often co-occurs with other forms of pollution, such as eutrophication and/or high levels of Polycyclic Aromatic Hydrocarbons (PAHs). Some of the challenges faced in modern environmental studies are similar to those encountered in deep-time, which uses geochemistry alone to understand environmental toxicity. Many studies that fail to find a relationship between ecological responses and heavy metals often overlook whether the heavy metals present in the environment were actually biologically available, e.g.,⁷². One example is the influx of Hg and other heavy metals into the Bay of Trieste over the last 300 years, where the lack of an ecological response in foraminifera was interpreted as evidence that the Hg was not biologically available⁷³, whereas molluscs from the same sediments show a clear ecological response, suggesting that Hg was indeed biologically available to certain taxa⁷⁴. This discrepancy highlights the importance of considering different ecological responses between phyla to heavy metal toxicity as well as the need for independent proxies for heavy metal bioavailability, such as heavy metal fractionation to evaluate the true ecological impact of heavy metal contamination.

Ecological responses beyond malformations

Ecological changes beyond malformations that could be associated with heavy metal toxicity for marine organisms have not been investigated for deep time events, with one exception (see Jiang et al.¹⁷). Expected community structural changes would be related to the different vital effects between species, where certain marine organisms are known to be more resilient to heavy metal toxicity than others, but also because heavy metals both bioaccumulate and biomagnify through the food chain. Hence, you may expect the selectivity of certain groups of organisms and organisms at higher trophic levels (Fig. 5). This means that compositional shifts within ecosystems/habitats, that can be quantitatively identified from the fossil record, could be related to heavy metal toxicity. Therefore, similar to modern day studies, one approach could be to investigate changes in the relative abundance of different species with changes in heavy metal bioavailability.

Another approach has been to investigate the incorporation of heavy metals in skeletal elements and animal tissues, e.g.,⁷⁵. Whilst it is obvious that animal tissues cannot be studied for past biodiversity crises, the incorporation of heavy metals into skeletal material could be informative. Caution, however, must be taken as the incorporation of heavy metals into the skeletal elements is physiologically controlled and not homogenous across an organism, and therefore the absence of heavy metals does not mean that the organism was not affected by heavy metal toxicity^{75,76}.

Biological response of terrestrial ecosystems

The impact of heavy metals on plant species is complex and varies widely based on factors such as the type of metal involved, concentration, exposure time, chemical form, soil composition, and pH levels^{77–79}. While some metals like iron (Fe), copper (Cu), zinc (Zn), nickel (Ni) and molybdenum (Mo) are bioessential micronutrients required for plant growth and development⁸⁰, their availability must be carefully regulated. Excessive concentrations of these metals can surpass the physiological need of plants, leading to toxic effects, including growth inhibition and cellular damage^{81–83}. In contrast, heavy metals such as lead (Pb), cadmium (Cd), and mercury (Hg), are non-bioessential and highly toxic, offering no benefits to plants and causing widespread disruption to vital physiological processes⁸⁴ (see Fig. 2). These metals interfere with crucial processes such as photosynthesis, nutrient uptake, and reproduction, often resulting in malformed sporomorph and/or reduced growth and survival^{82,83}.

Despite the different mechanisms by which plants uptake metals, the effects of lead (Pb), mercury (Hg), cadmium (Cd) and arsenic (As) share a common outcome: they induce oxidative stress. This stress is primarily driven by the generation of reactive oxygen species (ROS), which cause major damage to cellular structures and disrupt essential physiological processes. The accumulation of ROS in plants exposed to these heavy metals leads to oxidative damage in vital components like cell membranes, proteins, lipids, and DNA^{85,86}. This oxidative stress impairs vital functions like photosynthesis, nutrient and water uptake, and mitochondrial activity, all of which are crucial for the plant's growth and survival⁸⁷. Additionally, the breakdown of antioxidant defence mechanisms worsens this stress, making it difficult for plants to counterbalance the excess ROS^{88,89}. Consequently, plants suffer from stunted growth, chlorosis, reduced biomass, and compromised reproductive processes^{82,90,91}. In more severe cases, prolonged oxidative stress caused by heavy metal exposure can lead to genetic mutations, chromosomal abnormalities, and cell death, undermining the plant's ability to grow, reproduce, and maintain overall fitness^{92–94}. To mitigate these effects, plants employ both enzymatic and non-enzymatic pathways to scavenge ROS^{95,96}. In particular, metals such as Hg and Cd have been shown to interfere with key metabolic pathways by mimicking essential nutrients, thereby replacing them in enzymatic reactions and leading to metabolic dysfunctions⁹⁷.

Malformations in deep time

One of the few direct clues of biological stress preserved in the fossil record during mass extinction events is the ecological disruption observed within plant communities. These disruptions are often reflected as morphological changes in palynomorphs and leaf structures^{36,98–105}. Over the last decade, teratology, extreme morphological change or malformation, in terrestrial biota, has proven to be an excellent proxy for increased environmental perturbations. The strength of the method is that these deformities represent in vivo responses to environmental stress, and that they record this response as fossils. A well-documented example of such teratological features have been recorded at the Permian/Triassic boundary where widespread abnormalities in herbaceous lycopsids and *Isoetales* are highlighted by an increased abundance of unseparated *Isoetales*-tetrads and trisaccate pollen. These features have been linked to failures in microsporogenesis, indicative of impaired reproductive process^{106,107}. Initially, such morphological variations were attributed to enhanced solar ultraviolet B (UV-B) irradiation¹⁰⁸, presumably triggered by stratospheric ozone depletion associated with massive volcanic activity, which can damage plant DNA and interfere with normal cellular activity^{106,109–111}. However, emerging geochemical analysis have suggested that elevated concentrations of toxic metals, such as arsenic (As), cobalt (Co), mercury (Hg), and nickel (Ni), may have contributed to, or acted synergistically with, UV-B exposure in inducing mutagenesis in sporomorphs^{11,30,31,34,35,43,101,110–113}. Similar ecological disruptions and turnover in ecosystem composition are also well recorded at the Triassic/Jurassic boundary, characterized by a pronounced increase in malformed pollen and spores e.g.,^{99,100,105,114,115}. These morphological changes were interpreted as an

acute environmental stress, coinciding with intense volcanogenic activity known as the Central Atlantic Magmatic Province (CAMP).

Geochemical records reveal a positive shift in mercury concentrations across the boundary interval, temporally aligned with the rise in sporomorphs malformation. This correlation supports the hypothesis that the elevated atmospheric mercury emission during the CAMP volcanism may have been a key factor in inducing morphological changes within plant reproductive structure^{99,100,105}. Although less severe than the end-Permian or end-Triassic mass extinctions, the Toarcian event also exerted considerable stress on terrestrial vegetation. Sedimentary archives reveal spikes in mercury levels that align with extensive volcanic outgassing from the Karoo–Ferrar Large Igneous Province^{8,14,23}. This volcanism likely led to widespread atmospheric deposition of mercury and other heavy metals, contributing to the release of toxic compounds into the environment. Palynological records from this period show evidence of spore dwarfism, malformed spores, and asymmetrical *Classopollis* tetrads, often co-occurring with elevated concentrations of Hg, Cu, Cr, Cd, Pb, and As, indicative of prolonged environmental stress and physiological disruption in terrestrial plant communities³⁶.

In addition to morphological changes observed in spores and pollen, teratological features are not only confined to terrestrial plant communities but also extended to marine microplankton across extinction intervals, attributed to elevated levels of heavy metals. For instance, during the Late Ordovician extinction event Vandenbroucke et al.⁹⁸ and Munnecke et al.¹¹⁶ documented widespread teratology in chitinozoans and acritarchs accounting for up to 20% of assemblages. These morphological variations were strongly linked to heavy metal enrichment and marine acidification, likely resulting from extensive volcanic emissions.

Together, these studies underscore the use of fossil teratology as a sensitive and direct proxy for reconstructing episodes of environmental stress, particularly those involving volcanically driven perturbations in atmospheric and oceanic chemistry. However, teratological studies investigating environmental disturbances driven by non-volcanic sources—such as the Chicxulub impact associated with the end-Cretaceous mass extinction are still unknown.

Future directions/key outstanding questions

The hypothesis that heavy metal toxicity played a major role in deep-time biodiversity and mass extinction events is still largely an underexplored hypothesis that clearly requires further testing from multiple geoscientific perspectives.

In this review, we used the Permian Triassic mass extinction to highlight the complexities involved in evaluating heavy metal enrichments during hyperthermal events. First, the simple normalization of metal concentrations to a single baseline, such as TOC, may obscure the true environmental signal, as multiple geochemical processes can influence sedimentary metal accumulation. Second, lithological comparisons are challenged with potential misinterpretation, as differences in normalized values may reflect baseline variability rather than actual disparities in metal concentrations. Furthermore, even when metal enrichments are robustly identified, they do not automatically imply biological toxicity without assessing their bioavailability.

Key outstanding scientific questions include:

- Can heavy metal fractionation provide novel insights into the bioavailability of heavy metals during related extinctions?
- Were heavy metals biologically available at sufficiently high enough concentrations to be toxic and contribute to extinction during past biodiversity crises and mass extinction events?
- Are monstrosities and malformations in both the terrestrial and marine ecosystems related to metal toxicity or other environmental changes?
- Is heavy metal pollution at toxic levels a phenomenon exclusive to modern-day ecosystems or did it also contribute to extinction earlier in the Phanerozoic?

Future research should aim to refine toxicity thresholds by integrating diverse geological and biological data, expand spatial and temporal scales, account for species-specific responses, develop models that incorporate both nutrient deficiency and metal overaccumulation, and assess the influence of anthropogenic activities on heavy metal cycling and toxicity.

The questions outlined in this review paper highlight critical gaps in our understanding and addressing these will be essential for disentangling the complex environmental dynamics of Earth's deep-time history. By further investigating the bioavailability of heavy metals, their isotopic signatures, and their broader ecological impacts, we can gain invaluable insights into both past and present environmental challenges. We hope that the methods and models summarized herein will motivate further research and testing of these and other related questions.

Data availability

Data sharing not applicable to this article as no datasets were generated or analyzed during the current study.

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Author contributions

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Competing interests

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Additional information

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